

Interface Capability Brief

EDI 837 Claim Scrub — pre-clearinghouse rejection reduction · Stakeholder overview · 2026-08-13

1. Executive summary

SQL Server professional claims → payer-profile edits (referring NPI / Place of Service) → kickout work queue or clean 837 + SNIP before the clearinghouse. Every scrub decision is copied to a BI outcome folder.

2. Who this is for

- Sources: Edi837ClaimScrub (Claims PENDING) - Destinations: output/kickouts/, output/edi/, output/snip/, output/bi/ - Demo vs production: Synthetic payer rules mirrored in XSLT (not a live SQL join). SNIP on the clean path only.

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3. What the interface can do

Capabilities below are drawn from the working design and, when present, the published CapabilityStatement — not from aspirational roadmap items.

- Transport: DatabaseSqlListener (poll PENDING → PROCESSED)
- Format: SQLXML CLAIM rows (tags UPPERCASE)
- Identity: ClaimId / ClaimNumber
- Seed: 5001 clean AHLIC; 5002 kickout missing NPI; 5003 kickout POS 99; 5004 clean Medicaid POS 21

Integration highlights

FQCN sources: 835 PI (JDBC), 278 (XPath / FileWrite / NullTransport), this story's SNIP processor from PilotFish EDI SNIP docs. Image: pilotfish-eip:23R1.

Stage	Module / mechanism	Notes
R1 Listener	DatabaseSqlListener	SelectClaimsSQL.xml
R1 Transport	EIPTransport	Trigger Route 2
R2 Listener	TriggerableListener	Format Split Claims `//CLAIM \
R2 Evaluate	XSLTProcessor	Synthetic payer profile
R2 Gate	ExecuteProcessor OGNL on MatchBucket	kickout vs clean (no ; / colon-in-string)
R2 Clean	XSLT → EDITransformationProcessor XML→EDI 837-Q1 → SNIP 1–3	TableData mount
R2 Complete	NullTransport	

4. Honest boundaries (not claimed yet)

- None for this recreate.

5. What you can verify in a walkthrough

Use these as checklist items when reviewing the demo with business stakeholders. They describe outcomes, not module names.

- Exercise route “1 - Poll SQL Server Claims” from the Web UI or documented smoke path
- Confirm outputs land in the documented folders / queues and appear in the Web UI status views

6. How it works (plain language)

7. Validation

- Checked: kickouts 5002/5003; clean EDI+SNIP 5001/5004; SNIP HTML report - Not checked: live payer policy, SNIP 4–7

6. State & idempotency

- Claim marked PROCESSED on poll (before scrub completes) — demo pattern - Re-inject sets Status back to PENDING

8. Dual-write / side effects

Order: snapshot claims → evaluate → kickout and/or 837+SNIP → BI copy.

3. Inbound contract

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Routes in this interface

Each route is a configured PilotFish flow. Technical wiring appears in the Route Diagrams PDF; here is the stakeholder reading.

1 - Poll SQL Server Claims: runs steps such as Poll SQL Server Claims, Hand Off Claims.

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2 - Scrub Claim And Route: runs steps such as Scrub Claim And Route.Programmable (Trigger) Listener, Fork - XPath, No-Op Complete.

2 - Scrub Claim And Route: runs steps such as Scrub Claim And Route.Programmable (Trigger) Listener, Fork - XPath, Read Claim Keys, No-Op Complete.

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7. Security & trust posture

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- X12 SNIP validation is used where configured to reject bad EDI before downstream work
 - Risk noted in design: Med — Payer rules duplicated in XSLT — UI also reads PayerEditRules
 - Risk noted in design: Med — PROCESSED before scrub finishes — Demo-honest; re-inject resets
 - Risk noted in design: Low — SNIP heap — CATALINA_OPTS 6GB

8. Seeing it work

- - Ports: SQL 14341, EIP 8114, Web UI 8115 - SQL: sa / PilotFish_Demo1! · Edi837ClaimScrub - TableData: ../../../../EDI/TableData/x12 → edi-tabledata/837-Q1
- docker compose --profile full up -d --build
- | Web UI | http://127.0.0.1:8115/ |
- | EIP | http://127.0.0.1:8114/eip/ |
- | SQL | 127.0.0.1:14341 (sa / PilotFish_Demo1! · Edi837ClaimScrub) |

9. Related technical documents

- DESIGN.md — working engineering specification for this interface
- README.md — how to run, ports, and smoke commands
- This Capability Brief — shareable stakeholder summary (regenerate after design changes)

This document is generated from the interface sources listed above. If design and runtime diverge, believe the smoke-tested runtime and update DESIGN.md.